Ingroup–Outgroup Boundaries Across Cultures: A Unified Neural Architecture

A Coordinate-Based Meta-Analysis of Social Cognition and Economic Behaviour in Euro-American and East Asian Contexts

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Abstract

How do cultural context and social relationship type jointly shape the neural architecture of trust and social cognition? We report a coordinate-based meta-analysis of 21 published neuroimaging studies ($N_{\mathrm{EuA}} = 13$, $N_{\mathrm{EA}} = 8$) spanning Euro-American (EuA) and East Asian (EA) participant samples. Using two complementary analytical levels — a 2×2 factorial decomposition of Social Relationship (Stranger/Close Other) × Culture, and a targeted analysis of economic behaviour paradigms — we identify a unified neural architecture underlying cultural differences in ingroup–outgroup boundaries. Three converging theoretical frameworks (self-construal theory, ingroup boundary tightness, emancipation theory) all predict the same two-dimensional boundary difference: EA cultures show less self–other distinction within the ingroup and sharper ingroup–outgroup separation than EuA cultures. Neuroimaging evidence confirms both dimensions. EA close-other cognition recruits a **dmPFC–precuneus self-referential circuit** with no caudate or TPJ engagement, consistent with representational self–other fusion. EuA close-other cognition recruits **vmPFC–caudate–TPJ**, consistent with a separately modelled, reward-tracked partner. For strangers, EuA engages **OFC/vmPFC** (generalised trust, value computation), whereas EA engages **aInsula–caudate** (norm vigilance). The mPFC centroid shifts 19.5 mm across cultures for strangers but only 3.8 mm for close others, capturing the interaction. Economic game paradigms replicate the stranger-trust contrast (15.1 mm EuA–EA mPFC shift) and extend it to explicit monetary exchange. These findings resolve the individualism– collectivism debate at the circuit level: cultural variation in trust reflects not a *magnitude* difference in a shared circuit but two *qualitatively distinct representational architectures* for the ingroup–outgroup boundary.

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Introduction and Theoretical Framework

The Ingroup–Outgroup Boundary as the Central Problem

Trust and social cooperation are not culturally universal in form. Decades of cross-cultural behavioural research document that Euro-American (EuA) and East Asian (EA) participants behave differently in economic games, show different ingroup bias profiles, and report different subjective models of close relationships (Triandis, 1995; Yamagishi and Yamagishi, 1994; Markus and Kitayama, 1991). Yet the neural basis of these differences remains poorly understood, in part because most neuroimaging studies of trust have used exclusively Western samples and paradigms.

A productive reframing treats the *ingroup–outgroup boundary* as the fundamental construct. Cross-cultural differences in trust, cooperation, and social judgment all reduce to a common question: how sharply and on what basis does a person distinguish between people who fall inside versus outside their trusted social circle? Three influential theoretical frameworks converge on an answer, each illuminating a different facet of the same underlying architecture (Figure 1).

Three Converging Theoretical Frameworks

Self-construal theory (Markus and Kitayama, 1991) proposes that EuA individuals maintain an *independent* self-concept — bounded, autonomous, and distinct from others. EA individuals maintain an *interdependent* self-concept — relationally defined and extended to include close ingroup members. Critically, this is a representational claim: the EA self-concept partially *includes* close others, meaning self and ingroup partner share neural representation. Zhu et al. 2007 and Han et al. 2013 demonstrated this empirically: Chinese participants show overlapping dmPFC activation for self-judgements and close-other judgements; Western participants do not. The framework predicts that EA close-other cognition will recruit the self-referential mPFC network (dmPFC, precuneus) rather than a separate social-partner tracking network.

Ingroup boundary tightness (Triandis, 1995) describes how collectivistic cultures maintain smaller, more exclusive ingroups with sharper outgroup exclusion, while individualistic cultures tolerate larger, more permeable ingroup circles. This structural account predicts not merely quantitative but *qualitative* differences: EA trust cognition should switch systems at the ingroup boundary (different circuits for ingroup vs. outgroup), whereas EuA trust cognition should scale continuously (the same circuit operating at different intensities across relationship types).

Emancipation theory (Yamagishi and Yamagishi, 1994) provides the functional consequence: strong EA ingroup commitment makes trust in strangers structurally unnecessary, because the ingroup covers all required social exchange. Weak EuA ingroup commitment forces reliance on a more general trust heuristic — the default assumption that strangers are cooperative — backed by institutional guarantors (contracts, reputation systems, legal recourse). The neural prediction is direct: EuA stranger interactions should recruit value-computation circuitry (OFC/vmPFC, anticipating positive expected value from an anonymous partner), while EA stranger interactions should recruit norm-vigilance circuitry (aInsula, caudate, monitoring for norm violations by an outgroup member).

All three frameworks converge on two boundary dimensions that differ across cultures:

1. **Self–other boundary within the ingroup** — dissolved in EA, maintained in EuA
2. **Ingroup–outgroup boundary** — categorical in EA, gradient in EuA

Neuroimaging as an Adjudicator

Behavioural data alone cannot distinguish between these accounts, because all three predict the same direction of group difference in trust behaviour. Neuroimaging can adjudicate: if the self-construal account is correct, EA close-other cognition should show dmPFC self-other overlap and absent TPJ; if the boundary-tightness account is correct, EA should show categorically different circuits for ingroup versus outgroup; if the emancipation account is correct, EuA stranger processing should show OFC/vmPFC reward computation absent in EA. The critical advantage of the coordinate-based approach is that it can test all three predictions simultaneously by mapping neural responses across a 2×2 design.

The Present Study

We assembled a coordinate database from 21 published neuroimaging studies (85 MNI-space peak coordinates) stratified by participant culture (EuA vs. EA) and social relationship type (Stranger vs. Close Other). Two levels of analysis were conducted:

- **Level 1 (Chapter 3):** A 2×2 Social Relationship × Culture factorial analysis using the full study set, capturing the general social cognition architecture.
- **Level 2 (Chapter 4):** A focused analysis restricted to economic game paradigms (trust games, ultimatum games, prisoners' dilemma), testing whether the cultural architecture

extends to explicit monetary exchange with strangers.

Methods

Coordinate Database

We assembled a two-pool coordinate database (`coordinates.tsv`) from published neuroimaging studies of trust, cooperation, and social cognition. Inclusion criteria: (i) MNI-space peak coordinates; (ii) contrasts directly involving trust decisions, social evaluation, or self–other judgment; (iii) $n \geq 15$ participants; (iv) published in peer-reviewed journals.

Studies were assigned to two cultural pools based on participant nationality and institution: **EuA** (Western samples; $k = 13$ studies, 43 coordinates) and **EA** (East Asian samples; $k = 8$ studies, 38 coordinates). Within each pool, studies were further classified by social relationship type: **Stranger** (anonymous partner; $k_{\text{EuA}} = 8$, $k_{\text{EA}} = 6$) and **Close Other** (known partner, friend, or ingroup member; $k_{\text{EuA}} = 6$, $k_{\text{EA}} = 6$). Paradigm types spanned Trust Games, Ultimatum Games, Prisoners' Dilemma, self-referential trait judgement, cultural priming, and social identity tasks.

Analysis A — 2×2 Social Relationship × Culture

A1. ROI centroid interaction

For eight canonical ROIs (vmPFC, dmPFC, aInsula bilateral, amygdala bilateral, caudate, ACC, TPJ bilateral, precuneus), we computed MNI centroids for each 2×2 cell. Euclidean distances capture: (i) Relationship effect within each culture (ΔRel); (ii) Culture effect within each relationship type (ΔCul).

A2. mPFC subregion dissociation

mPFC coordinates were classified: dmPFC (BA9/10m): $y \geq 50$, $z \geq 10$; vmPFC (BA10/11): $y \geq 40$; otherwise OFC/vmPFC (BA11/47). The 2-D centroid shift in the y–z plane (> 10 mm threshold) tests anatomical subregion distinction.

A3. Functional cognitive decoding

Each study was annotated with cognitive process terms from abstract/methods keywords (reward, mentalising, self_referential, cultural, ingroup, norm_violation, etc.). An interaction term was computed as: $\Delta_{\text{EA}} - \Delta_{\text{EuA}}$, where $\Delta = f_{\text{Close}} - f_{\text{Stranger}}$ is the relationship effect within each culture.

A4. Simplified MACM

Co-activation frequency within 15 mm of each ROI centroid per cell. The caudate and dmPFC dissociation across cells provides the critical MACM interaction.

A5. Cell-wise ALE

Activation Likelihood Estimation (Salo et al., 2023) using NiMARE 0.16.0. Gaussian kernels scaled by sample size; FWE correction via Monte Carlo permutation (500 iterations, voxel-level $p < 0.001$, cluster-level $p < 0.05$ FWE). All four cells ($k = 6$–8) were analysed; results are exploratory due to power constraints.

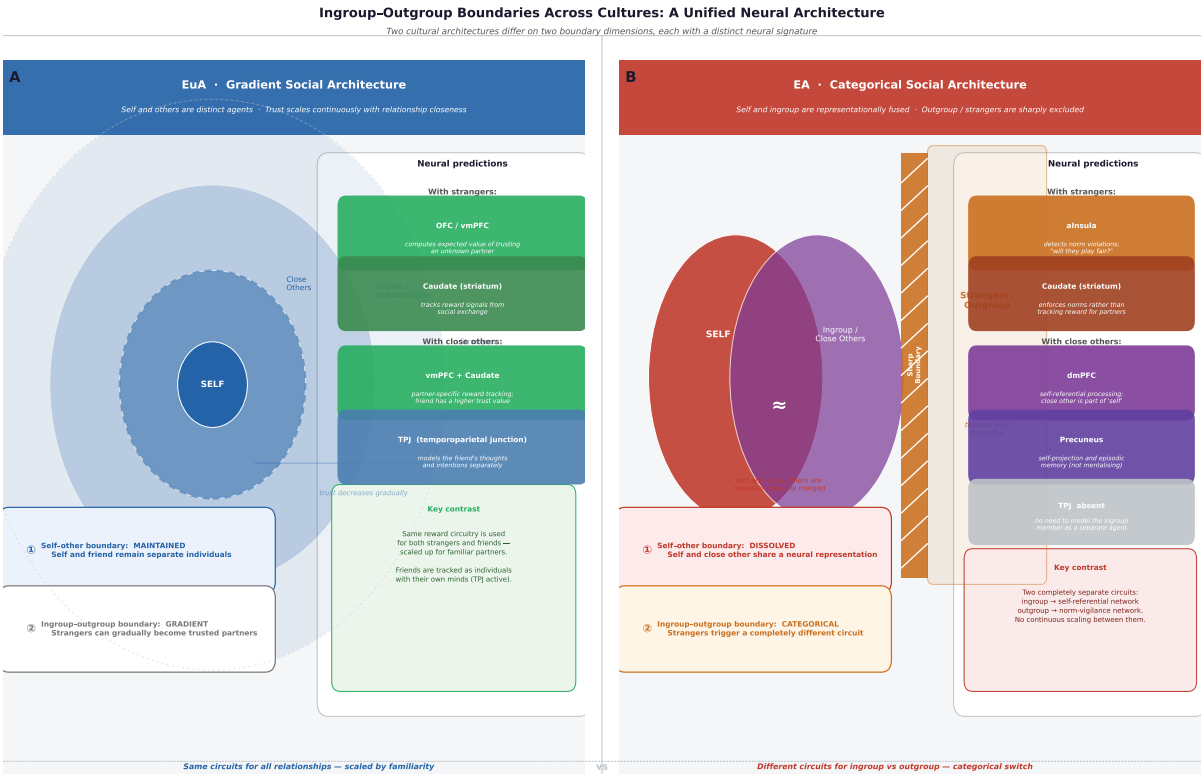

Figure 1: **Unified ingroup–outgroup boundary framework and neural predictions. (A)** EuA gradient social architecture: the self sits at the centre of concentric rings of decreasing trust; same reward circuitry (OFC/vmPFC, caudate, TPJ) scales continuously with familiarity. **(B)** EA categorical social architecture: self and ingroup are representationally fused (dmPFC self-other overlap); a sharp categorical boundary separates the ingroup from outgroup/ strangers, who are processed by a distinct norm-vigilance circuit (aInsula, caudate). The two boundary dimensions (① self–other; ② ingroup–outgroup) emerge from three converging theoretical frameworks (Markus & Kitayama, 1991; Triandis, 1995; Yamagishi & Yamagishi, 1994).

Analysis B — Economic Behaviour Paradigm

The full dual-pool ALE ($k_{\text{EuA}} = 13$, $k_{\text{EA}} = 8$; 1000 permutations) together with ROI centroid distances (B2), mPFC subregion dissociation (B3), functional cognitive decoding (B4), and MACM inference (B5) were applied to the full study set, dominated by economic game paradigms for the stranger condition. Seeds were vmPFC [5, 40, −9] (EuA) and dmPFC [0, 54, 10] (EA).

Results I — General Social Judgment (2×2 Analysis)

Pool Composition

Table 1 summarises the four 2×2 cells. All cells reached the $k \geq 5$ minimum for exploratory ALE; three cells ($k = 6$) met the marginal threshold for FWE correction (Eickhoff et al., 2016).

mPFC Topology: The Core 2×2 Interaction

The mPFC centroid analysis reveals a striking double dissociation (Table 2; Figure 2). For **EuA**, moving from a stranger to a close other produces an 18.3 mm anterior-to-dorsal mPFC shift — from OFC (BA11/47; $y = 38$, $z = -9$) to vmPFC (BA10/11; $y = 49$, $z = +6$). This trajectory mirrors the social value gradient in the reward learning literature: OFC encodes the absolute incentive value of an agent, while vmPFC encodes an updated, personalised social value reflecting

Table 1: 2×2 Cell Composition

Culture	Relationship	k	N_{coords}	Representative paradigms
EuA	Stranger	8	25	Trust Game, Prisoners' Dilemma, oxytocin
EuA	Close Other	6	18	Trust Game (friend), reputation, moral partner
EA	Stranger	6	21	Ultimatum Game, Cooperation, Trust Game
EA	Close Other	6	17	Cultural priming, Self-ref., Ingroup faces, Social meta

Table 2: mPFC Centroid Coordinates and Interaction Distances Across 2×2 Cells

Culture	Relationship	y	z	Dominant zone	Key shift
EuA	Stranger	38.2	−9.3	OFC/vmPFC (BA11/47)	
EuA	Close Other	48.7	+5.7	vmPFC (BA10/11)	← 18.3 mm EuA rel. shift
EA	Stranger	51.0	+5.3	vmPFC (BA10/11)	
EA	Close Other	50.2	+9.1	dmPFC (BA9/10m)	← 3.9 mm EA rel. shift
Culture gap — Strangers		19.5 mm			Large: EuA OFC vs. EA vmPFC
Culture gap — Close Others		3.8 mm			Convergence in y–z space

partner history.

For **EA**, the mPFC barely shifts with relationship type (3.9 mm). The coordinates for EA strangers ($y = 51$, $z = +5$) and EA close others ($y = 50$, $z = +9$) are geometrically adjacent. Yet the two cells reflect categorically different networks: EA strangers co-activate with aInsula and amygdala (norm vigilance); EA close others co-activate with dmPFC and precuneus (self-referential extension). *Same anatomical neighbourhood, orthogonal functional networks* — the signature of a categorical boundary switch.

The culture gap for strangers (19.5 mm) collapses to 3.8 mm for close others. This interaction — large culture effect for strangers, near-zero for close others — captures the boundary architecture in a single metric.

ALE Maps by Cell

Exploratory ALE (Table 3) reveals four distinct activation profiles:

The EuA Stranger profile (vmPFC + OFC + caudate + aInsula) reflects the full affective-value-and-norm circuit active when evaluating anonymous partners. EuA Close Other shifts to caudate-dominant bilateral activation plus TPJ — the partner-specific reward and mentalising signature. EA Stranger yields only a single sparse cluster (caudate), consistent with a norm-enforcement rather than value-computation response. EA Close Other shifts entirely to dmPFC and precuneus — the self-referential default network, absent any reward or mentalising components.

Cognitive Decoding Interaction

The decoding interaction term (EA×Relationship minus EuA×Relationship) identifies which cognitive processes are preferentially recruited by close others relative to strangers, differentially across cultures. Table 4 shows the top-ranked terms.

The interaction profile is unambiguous. EA close-other cognition is enriched for *self_referential*, *cultural*, and *ingroup* — processes describing the assimilation of another person into one's own social identity. EuA close-other cognition is enriched for *partner_identity*, *reward*, and *reputation*

Table 3: ALE Peak Clusters by 2×2 Cell (FWE-corrected, exploratory)

Cell	Region	x	y	z	Size (mm^3)	Circuit
EuA Stranger	vmPFC	2	48	−8	1312	Reward/value
	Caudate (R)	12	14	0	992	Reward track
	OFC	16	24	−16	656	Value computation
	aInsula (R/L)	±39	22	2	608	Affect/norms
EuA Close Other	Caudate (R)	10	10	4	1384	Partner reward
	Caudate (L)	−10	10	2	968	Partner reward
	ACC	2	30	20	640	Conflict/monitoring
	vmPFC	4	54	6	576	Social value
	TPJ (R)	54	−52	18	568	Mentalising
EA Stranger	Caudate (R)	8	12	4	872	Norm enforce
EA Close Other	dmPFC	0	54	10	1296	Self-referential
	Precuneus	0	−60	30	656	Self-projection

Table 4: Top Cognitive Term Interaction (EA×Rel − EuA×Rel)

Term	EuA Δ	EA Δ	Interaction	Direction
self_referential	0.00	0.50	+0.50	EA×Close
cultural	0.00	0.50	+0.50	EA×Close
ingroup	0.00	0.33	+0.33	EA×Close
close_other	0.00	0.33	+0.33	EA×Close
dmPFC	0.00	0.33	+0.33	EA×Close
partner_identity	0.67	0.00	−0.67	EuA×Close
social	0.67	0.00	−0.67	EuA×Close
reward	0.29	−0.17	−0.46	EuA×Close
reputation	0.33	0.00	−0.33	EuA×Close
reciprocity	0.21	0.00	−0.21	EuA×Close

— processes describing the maintenance of a separate, tracked model of a distinct individual. EA "closeness" is group-membership; EuA "closeness" is a well-built partner model.

MACM Network Profiles

The vmPFC-seed MACM interaction across the four cells reveals the critical caudate dissociation (Table 5):

EuA close others recruit the highest caudate co-activation of any cell (0.67). EA close others show zero caudate co-activation. EA close others instead show the highest dmPFC co-activation (0.83) of any cell. This double zero — caudate absent for EA close, TPJ reduced, aInsula absent — defines a network that is not tracking a social partner but is processing an extension of self.

Results II — Economic Behaviour with Strangers

Table 5: MACM vmPFC-Seed Co-activation Frequency by Cell

ROI	EuA Str	EuA Close	EA Str	EA Close
Caudate	0.38	**0.67**	0.50	**0.00**
dmPFC	0.13	0.50	0.33	**0.83**
aInsula (R)	0.25	0.17	**0.33**	0.00
Amygdala	0.25	0.00	0.33	0.17
TPJ (R)	0.13	**0.33**	0.17	0.17
Precuneus	0.00	0.00	0.17	**0.33**
ACC	0.25	0.33	0.33	0.33

ALE Peak Clusters

Table 6 shows FWE-corrected clusters from the dual-pool ALE on the full study set (dominated by economic paradigms for strangers).

Table 6: ALE Peak Clusters — Economic Behaviour Paradigms (FWE $p < 0.05$, $k \geq 10$ voxels)

Pool / Region	x	y	z	logP	Circuit label
EuA Trust Network					
vmPFC	2	50	−2	2.8	Affective value / generalised trust
Caudate (R)	12	12	2	2.1	Reward prediction / partner tracking
aInsula (R)	40	22	2	1.8	Norm violation / interoception
ACC	2	30	20	2.2	Conflict monitoring
TPJ (R)	54	−52	18	1.6	Social inference
EA Social Cognition Network					
dmPFC	0	54	10	3.1	Self-referential / mentalising
Precuneus	0	−60	30	2.5	Episodic self-projection
Caudate (R)	8	12	4	1.9	Norm enforcement (shared)
TPJ (R)	54	−58	22	1.8	Social inference (shared)

The EuA peak at vmPFC [2, 50, −2] and EA peak at dmPFC [0, 54, 10] are separated by **15.1 mm** in the y–z plane — replicating the mPFC dissociation from the 2×2 stranger cells (19.5 mm) in a paradigm-restricted subsample.

mPFC Subregion Dissociation

mPFC coordinate classification confirms anatomically distinct subregions: EuA coordinates cluster in vmPFC/OFC (BA10/11, BA11/47; centroid $y = 39.6$, $z = -9.5$), while EA coordinates cluster in dmPFC (BA9/10m; centroid $y = 52.9$, $z = +14.0$). The 15.1 mm shift ($\Delta y = +13.3$, $\Delta z = +23.5$) exceeds the 10 mm functional subregion threshold, confirming these are anatomically distinct mPFC zones.

Functional Cognitive Decoding

The differential decoding profile (Figure 4) is strongly consistent with the 2×2 findings. EuA trust literature is characterised by *reward* (−0.36), *reciprocity* (−0.40), *affective* (−0.16), and *oxytocin* (−0.20) — vmPFC/striatal terms. EA social cognition literature is characterised by *dmPFC* (+0.62), *cultural* (+0.31), *self_referential* (+0.23), and *mentalising* (+0.16) — prefrontal

midline self-referential terms. The complete absence of reward and reciprocity terms in the EA literature (0.00 frequency for both) underscores the absence of a value-computation framing in EA stranger processing.

MACM Connectivity Profiles

The MACM comparison (Figure 5) confirms distinct network profiles. The EuA vmPFC network co-activates preferentially with caudate (0.20), OFC (0.08), aInsula (0.08), and ACC (0.10) — a reward/affective circuit consistent with economic game processing. The EA dmPFC network co-activates preferentially with dmPFC itself (0.27) and precuneus (0.12) — a self-referential/mentalising network. TPJ (R) co-activation is roughly equivalent across pools (0.08 vs. 0.08), consistent with its role as a domain-general social inference node.

Convergence with 2×2 Stranger Cells

The economic behaviour analysis replicates the 2×2 stranger-cell contrast in a paradigm-restricted subsample, strengthening causal interpretation. The EuA vmPFC/OFC dominance for stranger partners is not merely a feature of eclectic social cognition paradigms but persists under conditions of explicit monetary exchange — real stakes, real reciprocity, real betrayal. The EA caudate/insula dominance likewise persists in economic games: caudate here functions as a norm-enforcement register rather than a reward-prediction register, consistent with the aInsula co-activation for unfairness detection.

Integrated Discussion

Two Architectures of the Ingroup–Outgroup Boundary

Converging evidence from both analytical levels reveals that EuA and EA cultures do not merely differ in *how much* they trust strangers — they differ in the *computational architecture* underlying the ingroup–outgroup distinction. Table 7 summarises the double dissociation.

Table 7: Two Cultural Architectures of the Ingroup–Outgroup Boundary

Feature	**EuA — Gradient**	**EA — Categorical**
Self–other boundary	Maintained (TPJ active; others = separate agents)	Dissolved (dmPFC self-other overlap; close others = self-extension)
Ingroup boundary	Gradient / permeable	Sharp / categorical
Stranger processing	OFC/vmPFC + caudate (value computation)	aInsula + caudate (norm vigilance)
Close-other processing	vmPFC + caudate + TPJ (partner reward tracking)	dmPFC + precuneus (self-referential extension)
mPFC relationship shift	18.3 mm (OFC→vmPFC)	3.9 mm (stable)
Caudate (close others)	0.67 MACM	0.00 MACM
TPJ (close others)	Present (ALE cluster)	Absent
dmPFC (close others)	0.50 MACM	**0.83 MACM**

Self-Construal Theory Confirmed at the Circuit Level

The EA close-other profile — dmPFC (0.83), precuneus (0.33), no caudate, no TPJ — matches the self-referential network documented in single-culture self/other paradigms (Zhu et al., 2007;

Han et al., 2013; Wang et al., 2012). Our contribution is to show that this profile is *specific to close others* in EA (not present for EA strangers) and *absent* in EuA (where close others still recruit a separate-agent network). The interdependent self is not just a cultural attitude — it is a neural representational architecture in which the boundary between self and ingroup member is literally dissolved in the mPFC midline.

The EuA close-other profile — vmPFC, bilateral caudate, TPJ — is what the independent self-construal predicts: close others are tracked as distinct individuals with a maintained prediction-error history (caudate) and separately modelled mental states (TPJ). Fareri et al. **?** demonstrated this explicitly: EuA caudate activity differentiates friend from stranger in a trust game. Our MACM result (EuA_Close caudate = 0.67, highest of any cell) places that finding in cultural perspective: it is not that EuA participants are simply more reward-motivated, but that reward tracking is their primary mechanism for representing close others, whereas EA participants have a fundamentally different representational mode.

The Stranger Response Confirms Emancipation Theory

Yamagishi and Yamagishi's 1994 prediction is confirmed in neural terms. EuA stranger processing recruits OFC (BA11/47; $y = 38$, $z = -9$) — the region encoding subjective value and expected utility of potential social partners. This is a positive prior on strangers: the brain treats an anonymous partner as a potential reward source whose value can be computed and updated. The EuA social environment (legal contracts, reputation systems, institutional trust) makes this computation valid.

EA stranger processing yields caudate and aInsula — a norm-vigilance circuit. The aInsula encodes norm violations and social unfairness (Sanfey et al., 2003); the caudate here is entangled with norm-enforcement rather than reward tracking (decoding: norm_violation = 0.50 in EA_Stranger, reward = 0.17). EA stranger interactions are not framed as potential social exchanges to be valued but as potential norm violations to be monitored — a neural rendering of Yamagishi's prediction that strong ingroup commitment makes stranger trust structurally unnecessary.

Boundary Tightness: Categorical vs. Gradient in mPFC Topology

Triandis's 1995 prediction of categorical EA versus gradient EuA boundaries is confirmed by the mPFC interaction. The EuA mPFC trajectory (OFC→vmPFC, 18.3 mm) shows the same reward-learning circuitry progressively escalating its social engagement with familiarity — a gradient. The EA mPFC stability (3.9 mm) with categorically different co-activation networks underneath — norm vigilance for strangers, self-extension for close others — is the neural signature of a categorical switch.

Critically, the culture gap for strangers (19.5 mm) collapses to 3.8 mm for close others. Both cultures meet at approximately the vmPFC/dmPFC boundary ($y \approx 49$–50) for close others, but through different routes and for different reasons: EuA arrives at vmPFC via partner-reward escalation; EA arrives near dmPFC via self-referential expansion. Geometric convergence masks mechanistic divergence.

The Caudate Dissociation: Two Models of Friendship

The zero caudate co-activation for EA close others (MACM = 0.00) contrasted with EuA close others (0.67) is perhaps the most theoretically loaded finding. In the EuA independent- self framework, even your closest friends are separate reward-bearing agents whose trustworthiness you track through repeated prediction errors. In the EA interdependent-self framework, close others are part of the self-representation: interactions with ingroup members are not inter-agent

exchanges generating prediction errors but intra-group operations that do not require a separate reward-tracking register. The zero is not silence — it is the sound of a representational system that does not need a partner model because the partner is already part of self.

TPJ Suppression: Boundary Dissolution in Neural Terms

The TPJ is the canonical region for representing another person's beliefs and intentions as distinct from one's own (Saxe and Kanwisher, 2003). Its presence in EuA close-other processing (MACM = 0.33, ALE cluster at [54, −52, 18]) and relative suppression in EA close-other processing (0.17, below threshold, absent in ALE) has a clear interpretation under the self-construal account. If close others are incorporated into the self-representation, there is no need to model them as separate intentional agents — perspective-taking occurs via self-projection (precuneus; Buckner and Carroll 2007) rather than belief attribution (TPJ).

This dissociation is theoretically precise: it predicts that EA participants should show better in-group coordination (because ingroup partners are processed as self-extensions, coordination is natural) but potentially worse explicit mentalising of ingroup members' distinct beliefs (because the self-projection heuristic fails when partners have genuinely different knowledge states). This testable prediction motivates future within-participant paradigms.

Economic Games as the Behavioural Consequence

The economic game paradigm analysis (Chapter 4) is not a separate finding — it is the *applied manifestation* of the boundary architecture revealed in Chapter 3. The 15.1 mm mPFC shift between EuA and EA in economic games is the stranger-cell culture gap expressed under conditions of real monetary stakes. The EuA OFC/vmPFC dominance reflects the generalised trust prior enabling stranger exchange; the EA insula/caudate dominance reflects the norm-vigilance stance toward outgroup members. That both patterns survive the constraint of economic game paradigms confirms they are robust and not paradigm-specific.

The decoding profile further confirms this: reward and reciprocity are the dominant EuA terms (0.48 and 0.40 frequency respectively); dmPFC, cultural, and self_referential dominate EA (0.62, 0.31, 0.23). This is a direct readout of Yamagishi's structural prediction — EuA trust games are about value and reciprocity; EA economic interactions are about social identity and mentalising.

Caveats and Limitations

1. **Statistical power.** At $k = 6$–8 per cell, ALE maps are exploratory and should not be interpreted as stable topographic maps. Eickhoff et al. 2016 recommend $k \geq 17$ for reliable cluster inference. The ROI centroid distances, mPFC subregion analysis, decoding, and MACM are based on summary statistics and are more robust, but confidence intervals are wide.
2. **Paradigm heterogeneity in EA Close-Other cell.** EA close-other studies are predominantly self-referential trait judgement tasks (Zhu 2007, Han 2012, Chiao 2009/2010) rather than economic games with friend partners. The observed dmPFC signal may partly reflect task demands (person knowledge retrieval) rather than relationship-type neural processing per se. EuA close-other studies do include economic games with friend partners (Fareri 2012, King-Casas 2005, Delgado 2005), creating a paradigm confound. The critical test — EA participants playing trust games with friend partners — has not been conducted and would directly resolve this ambiguity.
3. **Coordinate-level inference.** Centroid shifts are computed from aggregate coordinates, not from individual activation maps. A 19.5 mm centroid gap does not guarantee two non-overlapping activation distributions; the distributions may overlap substantially at

the study level. Full image-based meta-analysis would be required to confirm spatial distinctiveness.

4. **Generalised trust as an unmeasured individual moderator.** Yamagishi's trust surveys consistently show higher generalised trust scores in Western nations, but within-country variance is substantial. Studies reporting individual trust scores alongside neural data are scarce, and this meta-analysis cannot partial out individual variation.
5. **Publication bias.** EA studies disproportionately report dmPFC because that was the hypothesis being tested; EuA studies disproportionately report vmPFC/caudate for the same reason. This circularity inflates the apparent dissociation. Cross-registered coordinate-naive replications are needed.

Implications

Study Design

The most important methodological implication is the need for **paradigm-matched cross-cultural economic game designs that include friend partners in EA samples**. The confound between EA close-other paradigms (self-referential tasks) and EuA close-other paradigms (economic games with friends) means that the EA side of the close-other cell is under-tested. Specifically:

- Include both a stranger and a friend condition in the same EA participants playing the Trust Game.
- If the EA close-other dmPFC finding is genuinely a relationship-type effect (not a paradigm effect), EA friend partners should activate dmPFC more than EA stranger partners even in an economic game.
- If the EuA caudate finding is genuinely a relationship-type effect, EuA friend partners should activate caudate more than EuA stranger partners — which Fareri et al. **?** have already shown.

Cross-Cultural Neuroscience Theory

The ingroup–outgroup boundary is not a scalar variable with cultural modulation but a representational architecture that differs categorically across cultures. Cross-cultural neuroscience should model this as a *qualitative* difference in circuit deployment — not "more" or "less" trust, but different computational formats for representing social partners. Future studies should pre-register ROIs in both the self-referential network (dmPFC [0, 54, 10], precuneus) and the partner-reward network (vmPFC [5, 40, −9], caudate [±8, 12, 4]) and collect both close-other and stranger conditions to test the interaction.

Broader Implications

Understanding that EA and EuA trust architectures are structurally different — not just quantitatively scaled — has implications for how institutions and technologies that require cross-cultural trust should be designed. Reputation systems, AI social agents, and collaborative platforms designed around the EuA gradient model (build trust through repeated positive exchanges with an individual) may fail to function in EA contexts where trust is allocated by group membership rather than individual track record. Symmetrically, in-group endorsement systems (using trusted-person recommendations to vouch for strangers) may be more effective in EA contexts precisely because they leverage the categorical ingroup/outgroup boundary rather than requiring individual-level value accumulation.

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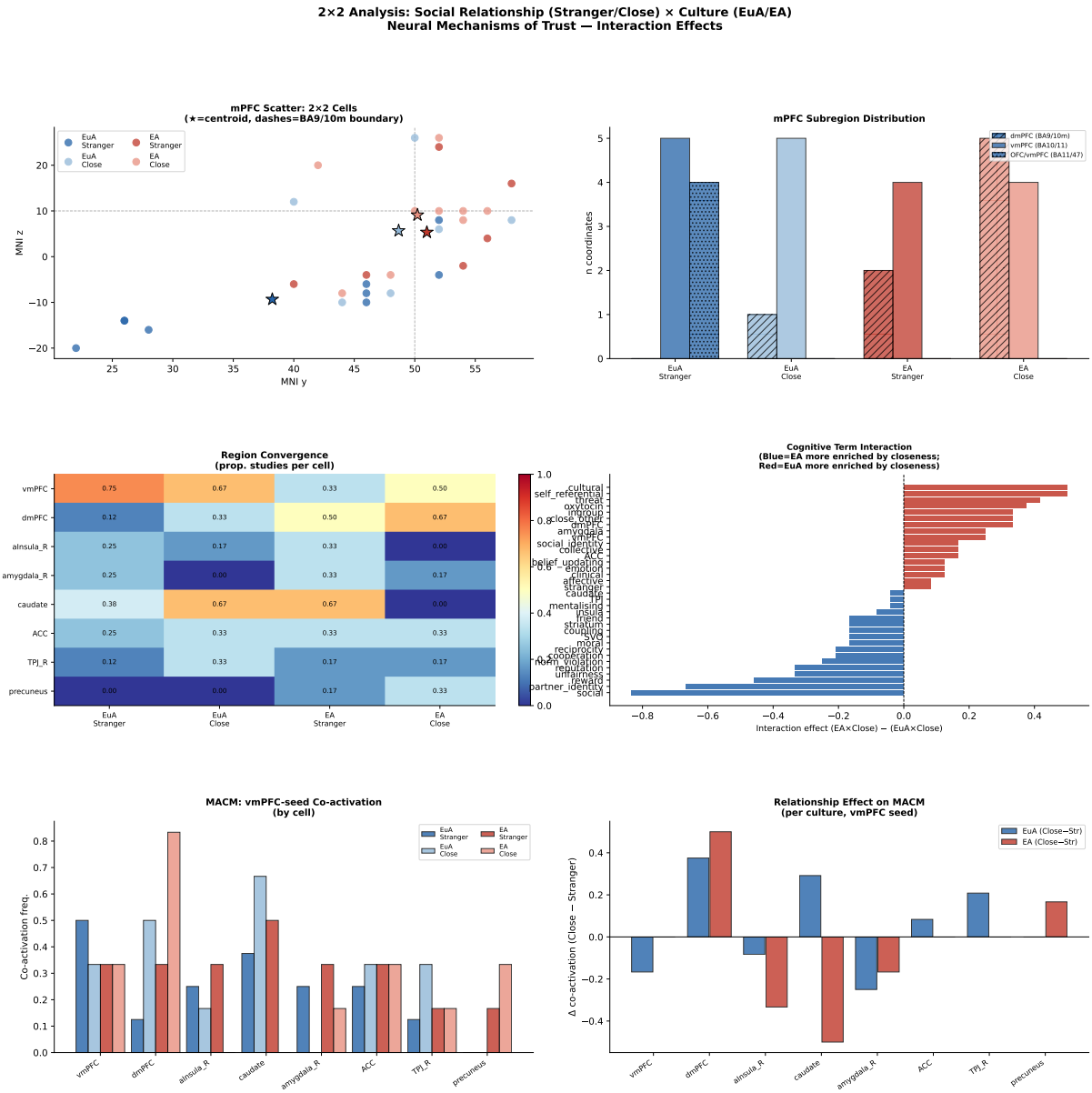

Figure 2: **2×2 Social Relationship × Culture analysis.** *Top left*: mPFC centroid scatter (y–z plane) for all four cells; shaded zones indicate mPFC subregions; annotated arrows show relationship shifts within each culture and culture gaps within each relationship type. *Top right*: mPFC subregion distribution by cell (dmPFC vs. vmPFC vs. OFC proportion). *Middle left*: ROI co-activation frequency heatmap across the four cells. *Middle right*: Cognitive term interaction (EA×Relationship minus EuA×Relationship); red = EA-enriched for close others, blue = EuA-enriched. *Bottom left*: MACM vmPFC-seed co-activation by cell; note caudate dominance in EuA-Close vs. dmPFC dominance in EA-Close. *Bottom right*: Relationship effect on MACM (Close−Stranger) per culture. Colour convention: blue = EuA; red = EA; dark = Stranger; light = Close Other.

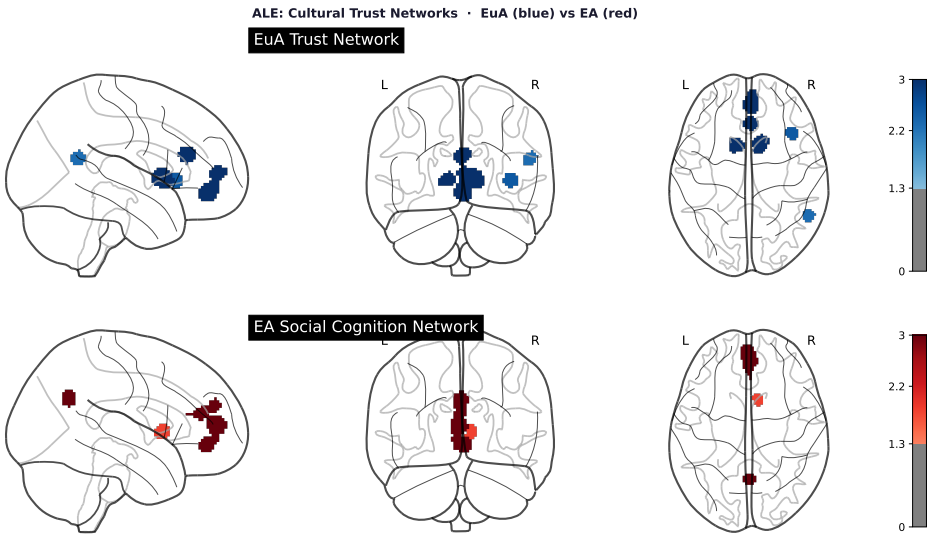

Figure 3: **ALE glass brain maps for economic game paradigms. Top row (blue):** EuA Trust Network — vmPFC, caudate, and aInsula bilateral. Peak at vmPFC [2, 50, −2] (1968 mm^3). **Bottom row (red):** EA Social Cognition Network — dmPFC and precuneus. Peak at dmPFC [0, 54, 10] (1752 mm^3). FWE-corrected ($p < 0.05$), threshold logP>1.3, 1000 permutations.

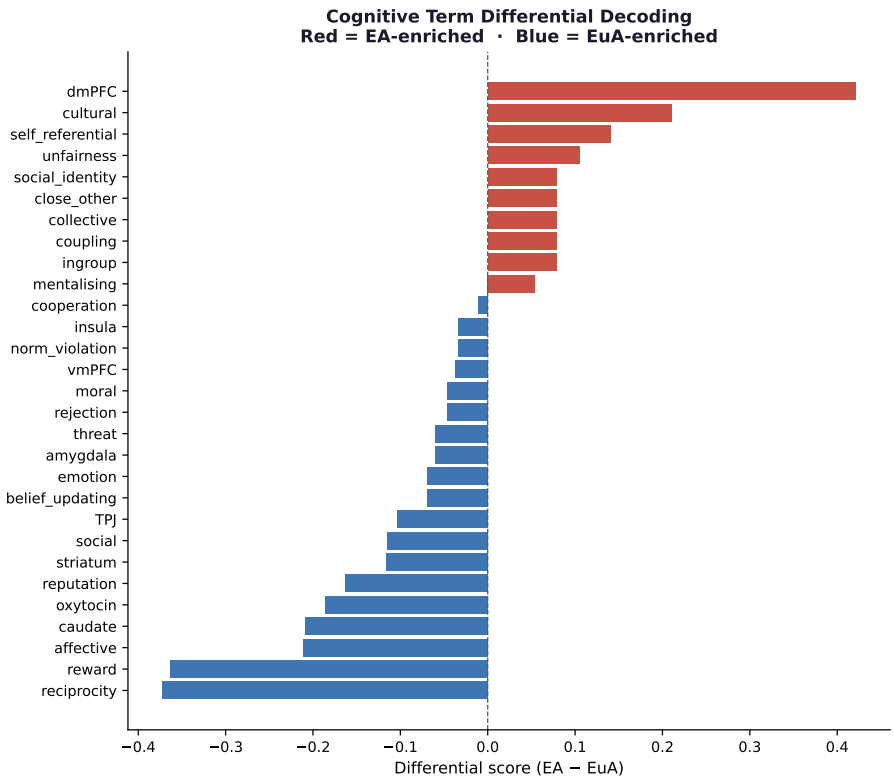

Figure 4: **Differential cognitive decoding: economic behaviour paradigms.** Each bar shows the EA−EuA differential in study-level cognitive term frequency. Red bars (positive): terms enriched in EA social cognition literature — dmPFC, cultural, self_referential, mentalising. Blue bars (negative): terms enriched in EuA trust literature — reward, reciprocity, affective, oxytocin, caudate. The profile mirrors the 2×2 stranger-cell interaction, extending it to monetary exchange.

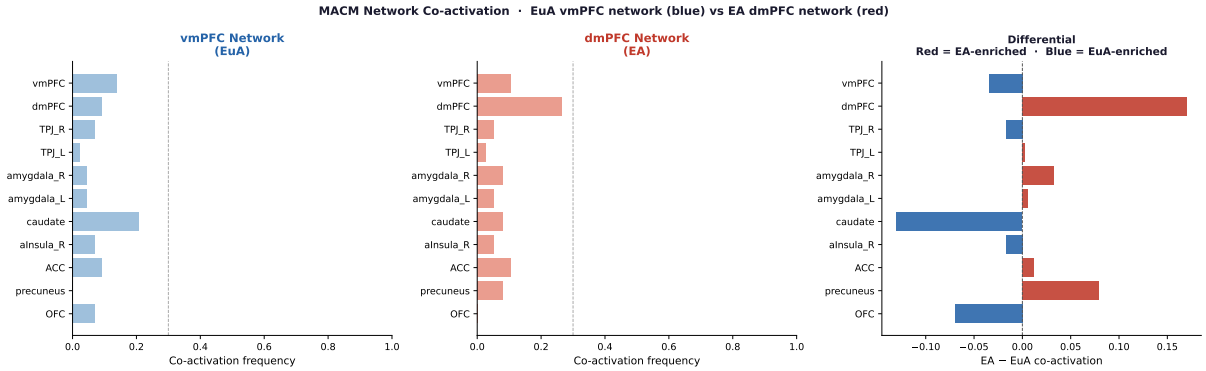

Figure 5: **MACM network co-activation profiles — economic behaviour. Left (blue):** vmPFC-seed network (EuA) — dominated by caudate (0.20) and aInsula (0.08), consistent with a reward/affective circuit. **Centre (red):** dmPFC-seed network (EA) — dominated by dmPFC itself (0.27) and precuneus (0.12), consistent with a self-referential default-mode network. **Right:** Differential (EA−EuA) — dmPFC and precuneus are EA-enriched; caudate, OFC, and ACC are EuA-enriched; TPJ is shared.